

Data Analytics Internship Portfolio

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Track: Data Analytics Internship @ DecodeLabs

Week 1

Project 1: Data Cleaning & Preprocessing

Microsoft Excel

Data Sanitization

Structural Formatting

The foundational phase of the internship focused on transforming raw, unstructured enterprise data into a reliable, analytical-ready dataset. The integrity of any predictive model or dashboard relies fundamentally on the hygiene of the input data.

1. Objective & Scope

To audit, clean, and standardize a messy sales dataset, removing inconsistencies, correcting data formats, and establishing a robust "single source of truth" for downstream analytical processes.

2. Execution Methodology

- **Deduplication:** Executed a comprehensive scan to identify and remove redundant row entries, ensuring transaction volumes and revenue metrics remain uninflated.
- **Missing Value Imputation:** Addressed null values logically. Missing nominal data was categorized appropriately, and numerical gaps were treated without skewing the variance.
- **Type Casting & Formatting:** Standardized temporal fields (Dates) and financial fields (Currency) into consistent formats, rectifying alignment and trailing/leading space issues.
- **Integrity Validation:** Conducted a final cross-reference check to ensure no anomalies persisted across the UnitPrice, Quantity, and TotalPrice columns.